Algebra 1 – Module 8 Practice Test •

WORKED KEY

Full Name: ______________________________

Period: _________

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Simplify: $\sqrt{80}$

Answer: $\sqrt{16 \cdot 5} = 4\sqrt{5}$

2. Simplify: $\sqrt{108}$

Answer: $\sqrt{36 \cdot 3} = 6\sqrt{3}$

3. Simplify: $\sqrt{64k^2}$

Answer: $8k$

4. Simplify: $\sqrt{6} \cdot \sqrt{3}$

Answer: $\sqrt{18} = 3\sqrt{2}$

5. Simplify: $5\sqrt{7} + 2\sqrt{7} - 3\sqrt{7}$

Answer: $4\sqrt{7}$

6. Simplify: $\sqrt{50} + 4\sqrt{2}$

Answer: $5\sqrt{2} + 4\sqrt{2} = 9\sqrt{2}$

7. Rationalize the denominator: $12/\sqrt{6}$

Answer: multiply by $\sqrt{6}/\sqrt{6}$: $12\sqrt{6}/6 = 2\sqrt{6}$

8. Rationalize the denominator: $4/\sqrt{20}$

Answer: $4/(2\sqrt{5}) = 2/\sqrt{5}$ → multiply by $\sqrt{5}/\sqrt{5}$: $2\sqrt{5}/5$

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9. Simplify: $(5n^3)^3$

Answer: $5^3 \cdot n^9 = 125n^9$

10. Simplify. Write your answer with a positive exponent: n^2 / n^8

Answer: $n^{-6} = 1/n^6$

11. Error Analysis — Rex evaluated $\sqrt{64 + 225}$ as $8 + 15 = 23$. Identify his mistake, then evaluate correctly.

Answer: the root of a SUM is not the sum of the roots — add first: $\sqrt{289} = 17$

12. Evaluate each expression.

a. 81^(1/2)

Answer: $\sqrt{81} = 9$

b. 216^(1/3)

Answer: $\sqrt[3]{216} = 6$

13. Solve: $\sqrt{x - 7} = 8$

Answer: square both sides: $x - 7 = 64 \rightarrow x = 71$

Cumulative Review — ACT Style. Circle the best answer.

14. Which ordered pair solves the system $x + y = 10$ and $x - y = 4$?

Ⓐ (6, 4)
Ⓑ **(7, 3)**
Ⓒ (3, 7)
Ⓓ (8, 2)

Answer: 7 + 3 = 10 ✓ and 7 − 3 = 4 ✓

15. Solve the compound inequality: $3 < x + 4 \le 9$

Ⓔ $7 < x \le 13$
Ⓕ $-1 \le x < 5$
Ⓖ $\mathbf{-1 < x \le 5}$
Ⓗ $3 < x \le 9$

Answer: subtract 4 from all parts

16. Which line is parallel to $y = -2x + 5$ through (0, −4)?

Ⓐ $y = 2x - 4$
Ⓑ $y = -(1/2)x - 4$
Ⓒ $\mathbf{y = -2x - 4}$
Ⓓ $y = -2x + 4$

Answer: parallel keeps slope −2; b = −4